CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech. (CL/EC/ME/ CE/EE/IT) SEM - II Examination, 2010

MA - 102 Engineering Mathematics II

Date: 21.12.10, Tuesday **Time: 10:30 a.m. to 01:30 p. m.** **Maximum Marks: 70**

Instructions:

1. **Attempt all the questions in each section.**
2. **Assume suitable additional data if required.**
3. **Figures to the right side indicate the full marks.**

SECTION-I

Q-1 (a) Solve: $\left[\cos x \tan y + \cos(x+y)\right] dx + [\sin x \sec^2 y + \cos(x+y)]\, dy = 0$ **[04]**

(b) Solve : $pq = 1$ **[04]**

(c) Solve: $pq = p + q$ **[03]**

Q-2 (a) Solve: $x^3 \dfrac{d^3y}{dx^3} + 2x^2 \dfrac{d^2y}{dx^2} + 2y = 10\left(x + \dfrac{1}{x}\right)$ **[04]**

(b) Solve: $\left(D^2 + 5D + 6\right) y = e^x$ **[04]**

(c) Solve: $z = px + qy + \sqrt{1 + p^2 + q^2}$ **[04]**

OR

Q-2 (a) Solve $\dfrac{d^2y}{dx^2} + y = \sec x$ by method of variation of parameters. **[04]**

(b) Solve: $x\dfrac{dy}{dx} + y = x^3 y^6$ **[04]**

(c) Solve: $\dfrac{dy}{dx} + \dfrac{y}{x} = x^3$ **[04]**

Q-3 (a) Solve the simultaneous linear equations : $\dfrac{dx}{dt} - 7x + y = 0$, $\dfrac{dy}{dt} - 2x - 5y = 0$ **[04]**

(b) Solve: $p^2 + q^2 = x^2 + y^2$ **[04]**

(c) Solve: $x^2 p + y^2 q = (x + y) z$ **[04]**

OR

Q-3 (a) Solve: $\left(y^2 + z^2\right) p - xyq - xz = 0$ **[04]**

(b) Solve: $(x+1)^2 \dfrac{d^2y}{dx^2} + (x+1)\dfrac{dy}{dx} + y = 2\cos[\log(1+x)]$ **[04]**

(c) Solve the simultaneous linear equations : $\dfrac{dx}{dt} + y = \sin t$, $\dfrac{dy}{dt} + x = \cos t$ **[04]**

[P.T.O]

SECTION-II

Q-4 (a) Show that $erf(x)+erf_c(x)=1$ [03]

(b) Evaluate: $\int_0^1\int_0^1 \frac{dx\,dy}{\sqrt{(1-x^2)(1-y^2)}}$ [03]

(c) Verify that the matrix $A=\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ satisfies the Cayley-Hamilton theorem and hence obtain A^{-1}. [05]

Q-5 (a) Prove that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} \times \int_0^{\pi/2} \sqrt{\sin\theta}\ d\theta = \pi$ [03]

(b) If $A=\begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, Show that (i) A is Hermitian (ii) iA is Skew-Hermitian [05]

(c) Find the area between parabolas $y^2=4ax$ and $x^2=4ay$. [04]

OR

Q-5 (a) Prove that $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4}B\left(\frac{1}{4},\frac{1}{2}\right)$ [03]

(b) Prove that $\frac{1}{2}\begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is a unitary matrix. [05]

(c) Find the area lying between the parabola $y=4x-x^2$ and the line $y=x$. [04]

Q-6 (a) For Legendre Polynomials $P_m(x), P_n(x)$, show that $\int_{-1}^{1} P_m(x)P_n(x)\,dx=0$, if $m \neq n$. [04]

(b) Adopting standard notations, Show that $\int x J_0(x)\,dx = xJ_1(x)$ [04]

(c) Evaluate: $\int_0^1\int_0^{1-x}\int_0^{(x+y)^2} x\,dz\,dy\,dx$ [04]

OR

Q-6 (a) For integer n, show that $\int_{-1}^{1} P_n^2(x)\,dx = \frac{2}{2n+1}$, where $P_n(x)$ is a Legendre Polynomial. [04]

(b) Evaluate: $\int_0^1\int_0^1\int_0^1 e^{x+y+z}\,dx\,dy\,dz$. [04]

(c) Adopting standard notations, Show that $\int x^2 J_0(x)J_1(x)\,dx = \frac{x^2 J_1^2(x)}{2}$ [04]
